CHAPTER 18

Uses of Radioactive Tracers

Contents

18.1. Basic Assumptions for Tracer Use 546
18.2. Chemistry of Trace Concentrations 547
- 18.2.1. Adsorption 548
- 18.2.2. Radiocolloids 550
- 18.2.3. Equilibrium reactions 550
- 18.2.4. Precipitation and crystallization 550
- 18.2.5. Electrochemical properties 552
- 18.2.6. Tracer separation methods 553

18.3. Analytical Chemistry 555
- 18.3.1. Radiometric analysis 555
- 18.3.2. Isotope dilution analysis using radiotracers 556
- 18.3.3. Activation analysis 558
- 18.3.4. Substoichiometric analysis 562

18.4. Applications to General Chemistry 564
- 18.4.1. Determination of chemical reaction paths 564
- 18.4.2. Determination of chemical exchange rates 567
- 18.4.3. Determination of equilibrium constants 568
- 18.4.4. Studies of surfaces and reactions in solids 572

18.5. Applications to Life Sciences 573
- 18.5.1. Biological affinity 574
- 18.5.2. Transmission computer tomography (TCT) 577
- 18.5.3. Emission computer tomography (ECT) and diagnosis 579
- 18.5.4. Radiation therapy with internal radionuclides 586

18.6. Industrial Uses of Radiotracers 586
- 18.6.1. Mixing 586
- 18.6.2. Liquid volumes and flows 587
- 18.6.3. Wear and corrosion 588
- 18.6.4. Chemical processing 588

18.7. Environmental Applications 588
18.8. Exercises 590
18.9. Literature 592

In this chapter some of the ways in which radiochemistry has aided research in various areas of chemistry and related sciences are reviewed.

The first experiments with radioactive tracers were conducted in 1913 by de Hevesy and Paneth who determined the solubility of lead salts by using one of the naturally occurring radioactive isotopes of lead. Later, after discovery of induced radioactivity,

Radiochemistry and Nuclear Chemistry
ISBN 978-0-12-405897-2, http://dx.doi.org/10.1016/B978-0-12-405897-2.00018-5
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de Hevesy and Chiewitz in 1935 synthesized ^{32}P (β^- $t_{½}$ 14.3 d) and used this tracer in biological studies. In the same year de Hevesy and co-workers also carried out activation analyses on rare earths. Despite the demonstration of the value of the tracer technique by these early studies the technique did not come into common use until after World War II when relatively large amounts of cheap radionuclides became available through the use of nuclear reactors.

While it is not necessary to use radioactive isotopes for tracer studies, in general, the use of radioactivity is simpler and less expensive than the use of stable isotopes. Research with the latter requires rather sophisticated and expensive measuring devices such as mass spectrometers, cf. §3.3.2. We restrict our discussion to the use of radioactive tracers.

Among the advantages of using radiotracers we can list the following: (a) radiotracers are easy to detect and measure with high precision to sensitivities of 10^{-16} to 10^{-6} g; (b) the radioactivity is independent of pressure, temperature, chemical and physical state; (c) radiotracers do not affect the system and can be used in nondestructive techniques; (d) if the tracer is radiochemically pure, interference from other elements is of no concern (common in ordinary chemical analyses); (e) for most radioisotopes the radiation can be measured independently of the matrix, eliminating the need for calibration curves.

18.1. BASIC ASSUMPTIONS FOR TRACER USE

In some experiments answers to scientific questions which require knowledge of the presence and concentration of a specific element or compound at a certain place and at a certain time can be obtained only through the use of a radioactive tracer. For example, self diffusion of metal ions in solutions of their salts cannot easily be studied by any other technique. However, in other cases the use of radioactive tracers is not necessary in principle but is justified by the greater convenience. In either type of investigation there are two assumptions implicit in such uses.

The primary assumption is that radioactive isotopes are chemically identical with stable isotopes of the same element, i.e. the substitution of ^{14}C for ^{12}C in a compound of carbon does not change the type or strength of the chemical bonds nor does it affect the physical properties of the compound. The validity of this assumption depends on the precision of measurement of the chemical and physical properties. The difference in mass between the various isotopes does cause some change in these properties (§3.5) but even in the case of ^{14}C and ^{12}C, with a mass difference of approximately 15%, the isotope effect is rather small and difficult to detect. Normally only for systems involving hydrogen-deuterium-tritium substitution must isotope effects be considered. For heavier elements it can be neglected in almost every situation.

The second assumption in the use of tracer techniques is that the radioactive nature of the isotope does not change the chemical and physical properties. Until the moment of